Sahil Mukesh Deo

B.Tech, Computer Science and Engineering
Indian Institute of Technology Bombay
Minor in Statistics • Graduating May 2029

Ph: +91-7498046893
Email: 25b0991@iitb.ac.in
Email: sahildeo54@gmail.com
LinkedIn: sahil-mukesh-deo
GitHub: Sahil-M-Deo
Codeforces: Sahil_Deo

Examination	Institute	Year	CPI/%
Graduation	Indian Institute of Technology Bombay	2025 – present	9.68
Intermediate	Narayana e-Techno School	2025	96.8%
Matriculation	Wisdom High International School	2023	99.2%

Scholastic Achievements

- Secured **All India Rank 7** in **JEE Advanced** among 180,000+ candidates *2025*
- Received the **Institute Academic Award** for securing **Rank 25** among 1,360 students by CPI *2026*
- Secured **Department Rank 15** among 200 students in CSE, IIT Bombay, by CPI *2026*
- Reached the **IICPC QuantFest** regionals (**top 600** of 10,000+); **Rank 227** at the regionals *2026*
- Achieved **All India Rank 102** in **JEE Main** among 1,400,000+ candidates *2025*
- Selected for the **Indian National Physics Olympiad** (**top 300** of 40,000+ in NSEP) *2024-25*
- Qualified for the **Indian National Astronomy** and **Chemistry Olympiads** (INAO, INChO) *2024-25*
- Secured **1st position** in **Physics Bazinga** and in **Maths Bazinga**, Maths & Physics Club *2025*
- Rated **1654** on Codeforces *2026*
- Shortlisted for the **Aditya Birla Scholarship** interview *2026*

Professional Experience

First-Year Trading and Technology Program (FTTP) | *Jane Street, Hong Kong* *(2026)*

- Shortlisted for **Jane Street's** selective program for first-year STEM undergraduates
- Attended classes on **trading and technology** and competed in a **team-based mock trading simulation**
- Developed **fair-value estimates** for a complex **live market simulation** with market inefficiencies

Key Projects

Randomised Algorithms: A 137-Page Book | *Summer of Science* *(2026)*
Maths and Physics Club, IIT Bombay

- Wrote a **137-page book** on randomised algorithms, from **hashing** to **Pollard's rho** and **HyperLogLog**
- **Conjectured** from simulation (exhaustive for small n, **Monte Carlo** for large n) that the 5-cycle is the worst case for a randomised **max-cut** algorithm, which would cut its expected time from $\Theta(nm)$ to $\Theta(n+m)$
- Proved that n pairwise-independent bits need $\geq \log_2(n+1)$ truly random bits (tight)

Game Theory | *Learners' Space*
Maths and Physics Club, IIT Bombay

- Studied **mixed Nash equilibria** and the **minimax theorem**; solved zero-sum games as **LPs**
- In the n-player **volunteer's dilemma**, showed P(nobody volunteers) $\to$ **cost/benefit**, not 0, as $n \to \infty$

Deep Chess | *Seasons of Code*
Web and Coding Club, IIT Bombay

- Built a C++ chess engine rated **2200** with **alpha-beta search** and a **Zobrist-hashed transposition table**
- Trained an **NNUE evaluation network** in **PyTorch** with incremental updates for fast inference

Positions of Responsibility

- Taught a **Discrete Structures** tutorial (**Tutorial Services Centre**) with **original problems** *2026*

Skills & Courses

Languages & Tools C++, Python, Java, JavaScript, Bash, LaTeX; Git, NumPy, PyTorch
Courses Data Analysis & Interpretation, Data Structures & Algorithms, Discrete Structures